

Nexus 360Web API Documentation

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Token Access

The token API is used to retrieve access token required for accessing protected Nexus360 resources. The Bearer Token MUST be included in the Header to access any Nexus360 APIs

Response

Status: 200 OK

Body:

```
{
  "access_token": "BDXRDeA_-
niWhnMgb1g8gafosqPxdbDwRpnnWCV8Pw5SYLfC7ljow7Ay_mEIWD2G0KYIdtNchxWao6H7
3AR_o0IIJWcQ9TYb0maNn_bXP0R8U5oJIZ489pX8nbuJ4_dCX0Xhf20dUQ6d0CDzVd_6ELa-
GFfXZL05CG4RZSV3cQ8v_0Im.....netW9PgZ-4Cf-9wPuKsUk1uP076-
0RsgR1uJ1o5aI8tPrwJ00h4jADSE",
  "token_type": "bearer",
  "expires_in": 86399
}
```

Example Request

```
curl \
--header "Content-Type:application/x-www-form-urlencoded" \
--request GET \
--data "grant_type=password&username=devicekey&password=#####" \
https://batchtest.nexus360.cloud/token
```

```
{"access_token":"QGhJBXlKZ380uCLcnUUjYZ4QqDeaX07x0m14tlt2n-
ElBgnTHirXBNNbQsv-Kthj4C2GX0nyunlxlYdRoA1qPN-kH__Y6e9k-
IvK3SI6TEeoffoH30N3x0Z0sxUGe4REahso-
5bqroxjeqn7tiC5IzUTQh06DH4e2xLcD9nKuV2y0dn0T0ffwapWdTTt4SwZephcmNw9udCvo
aVfxypoplmaA35MPaPx6jDt0tDExLj00pokgX0mAJ0VA49AVNX5gFicCi5glBz8Ml4JTpY0eC
RSQExek1J4PvwT-mtwga21A", "token_type":"bearer", "expires_in":86399}
```

Web APIs

End Point: `api/getservicelistforsite/{id}`

Remarks:

HTTP Method: GET

Parameters: string id

Description: Retrieves a list of services associated with a specific site. The list is obtained by calling a stored procedure that requires a user and site ID.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- id: A string representing the unique identifier of the site. This ID is used to retrieve the list of services associated with the specified site.

Returns:

- Ok - response with a list of objects if the request is successful.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

```
[
{
"ServiceID": "example-service-id-1",
"ServiceName": "Example Service 1",
"Description": "Description of Example Service 1",
"SiteID": "D625BFD0-4D0F-4F88-A2FD-ED86339B77AC"
},
{
"ServiceID": "example-service-id-2",
"ServiceName": "Example Service 2",
"Description": "Description of Example Service 2",
"SiteID": "D625BFD0-4D0F-4F88-A2FD-ED86339B77AC"
}
]
```

End Point: `api/getpatientid/{id}`

Remarks:

HTTP Method: POST

Parameters: [FromBody] Model

Description: Retrieves the unique identifier (ID) of a patient based on the provided patient details such as reference, last name, and first name. The method ensures that the user has the necessary permissions before processing the request.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- pat: An object of type SimplePatientModel containing the following properties:

Example of [FromBody] Model:

```
{
    public string reference { get; set; }
    public string lastName { get; set; }
    public string firstName { get; set; }
    public string sex { get; set; }
    public string dateOfBirth { get; set; }
}
```

Returns:

- Ok - response with the patient ID as a string if the request is successful and the patient is found.
- Forbidden - response if the user does not have the necessary permissions to view the patient information.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

```
{
  "PatientID": "example-patient-id"
}
```

End Point: `api/findpatient/SearchLastname=" + {searchdata}`

Remarks:

HTTP Method: GET

Description: Searches for a patient based on the provided search data. The method ensures that the user has the necessary permissions before processing the request.

Examples of : "ID=s": Finds a patient by ID. "SearchLastname=s": Finds patients with a similar last name. "SearchFirstname=s": Finds patients with a similar first name. "SearchReference=s": Finds patients with a similar reference. Note: If more than one search criterion is provided, only the first one will be used.

Parameters:

- searchdata: The search criteria string.

Returns:

Returns: List of Patient object that fit within the search parameters

Example:

///

Patient Object Example:

```
{  
patientID = Guid "123e4567-e89b-12d3-a456-426614174000",  
clusterID = Guid "abcdefab-abcd-abcd-abcd-abcdefabcdef",  
reference = "ABC123",  
creationDate = 20200101,  
socialSecurityNo = "123-45-6789",  
title = "Mr.",  
lastName = "Doe",  
firstName = "John",  
middleName = "Michael",  
dateOfBirth = 19800101,  
dateOfDeath = null,  
conceptionDate = null,  
sex = "Male",  
race = "Caucasian",  
religion = "None",  
maritalStatus = "Single",  
primaryLanguage = "English",  
birthPlace = "Springfield",  
nationality = "American",  
occupation = "Engineer",  
patientGroup = "General",  
medicareNumber = "123456789",  
medicareIndex = "1",  
pensionNumber = "987654321",  
safetyNetNumber = "555",  
vetAffairsNumber = "666",  
socialSecurityNo = "123-45-6789",  
street = "123 Elm Street",  
city = "Springfield",  
region = "IL",  
postCode = "62701",  
county = "Sangamon",
```

```
country = "USA",
homePhone = "(555) 123-4567",
workPhone = "(555) 234-5678",
mobilePhone = "(555) 345-6789",
email = "john.doe@example.com",
nextOfKin = "Jane Doe",
emergencyContact = "Jane Doe, (555) 456-7890",
emergencyPhone = "(555) 456-7890",
emergencyContact2 = "Michael Doe",
emergencyPhone2 = "(555) 567-8901",
bloodGroup = "O+",
smokerType = "Non-smoker",
drinkerType = "Social",
allergies = "Penicillin",
notes = "No known allergies. Regular check-ups required.",
deleteDate = new DateTime(9999, 12, 31),
version = 1
};
```

End Point: `api/getstudystate/{id}`

Remarks:

HTTP Method: GET

Parameters: string id

Description: Retrieves the state of a study based on the provided study ID. This method returns the current status of the study if it exists.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- id: The unique identifier (ID) of the study. This should be a valid GUID string.

Returns:

- Ok - response with the study state as a string if the study is found and the request is successful.
- Unauthorized - response if the authorization header is missing or invalid.
- NotFound - response if no study is found for the provided ID.
- InternalServerError - response if an unexpected error occurs during processing.

Exceptions:

- T:System.Exception: Logs and rethrows any exceptions encountered during data retrieval.

Example:

Example of a successful response:

"Device Configured"

End Point: `api/setstudystate`

Remarks:

HTTP Method: POST

Parameters: FromBody Model

Description: Updates the state of a study based on the provided study ID and new state information. This method checks for specific conditions, such as whether the study is an EEG study, to determine the appropriate state.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- info: An object of type StudyStateInfo containing the following parameters:

Example of request Body:

```
{  
  "studyID": "1234abcd-5678-efgh-9101-1121ijklmno",  
  "studyState": "Uploaded",  
  "docID": "abcd1234-5678-efgh-9101-ijklmnop1234",  
  "application": "App"  
}
```

Returns:

- OK - response if the study state is successfully updated.
- Forbidden - response if the user does not have the necessary permissions to update the study state.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

```
{  
  "status": "OK",  
  "message": "Study state updated successfully."  
}
```

End Point: `api/CreateStudy`

Remarks:

HTTP Method: POST

Parameters: FromBody Model

Description: Creates a new study record in the system based on the provided details. This endpoint handles the creation of both the study and related documents, and sets up the necessary file paths and study metadata.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- info: An object of type CreateStudyInfo containing the details of the study to be created. This includes the following fields:

Example of a request body:

```
{  
  "workstationID": "e9c6b7f3-df6a-4b29-bc6a-8b10c91b04e4",  
  "patientID": "d3b0a8e3-c5b5-4c93-9d52-6cbb61685b56",  
  "serviceID": "c7d1f63a-b6f4-4c32-bf2f-382bd7ea0bb0",  
  "siteID": "e8a1a5c6-bb65-4dcb-9c0e-1d18d6377e80",  
  "studyStart": "9/12/2024 3:30:00 PM",  
  "studyState": "Active",  
  "localCachePath": "C:\\Studies\\Study123",  
  "notes": "This is a test study.",  
  "weight": "70",  
  "height": "180"  
}
```

Returns:

- Ok - response containing the document ID and study ID if the study was successfully created.
- Forbidden - response if the user does not have the necessary permissions to create a study.
- InternalServerError - response if an unexpected error occurs during processing, including errors related to workstation ID, document creation, or file system issues.

End Point: `api/getpatientinfostring/{id}`

Remarks:

HTTP Method: GET

Parameters: string id

Description: Retrieves patient information as a string based on the provided patient ID. This method extracts and formats the patient information into a readable string format.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- id: The unique identifier (ID) of the patient. This should be a valid GUID string.

Returns:

- Ok - response with a string containing patient information if the request is successful and the patient ID is valid.
- Unauthorized - response if the authorization header is missing or invalid.
- NotFound - response if no patient is found for the provided ID.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful response:

"John Doe (CMP0001)"

End Point: `api/getserviceattributes`**Remarks:**

HTTP Method: GET

Parameters: string id

Description: Retrieves the attributes of a specified service based on the provided service ID. This method checks the user's role to ensure they have the necessary permissions to view service attributes.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- id: A string representing the unique identifier of the service. This should be a valid GUID.

Returns:

- Ok - response with a list of service attributes if the request is successful and the user is authorized.
- Unauthorized - response if the user does not have the necessary permissions to view the service attributes.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

[

```
{
  "AttributeID": "example-attribute-id",
  "AttributeName": "AttributeNameExample",
  "AttributeValue": "AttributeValueExample"
},
{
  "AttributeID": "example-attribute-id-2",
  "AttributeName": "AttributeNameExample2",
  "AttributeValue": "AttributeValueExample2"
}
]
```

End Point: `api/CreatePatient`

Remarks:

HTTP Method: POST

Parameters: FromBody Model

Description: Creates a new patient record in the system using the provided patient details. This endpoint processes the patient information and adds a new entry to the database.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- : An object of type `apiPatient` containing the details of the patient to be created. This includes the following fields:

Example of a request FromBody:

```
{
  "reference": "PAT123456",
  "lastName": "Doe",
  "firstName": "Jane",
  "middleName": "A.",
  "title": "Ms.",
  "dateOfBirth": "1985-02-15",
  "sex": "Female",
  "street": "456 Elm St",
  "city": "Anytown",
  "region": "AnyState",
  "postCode": "12345",
  "country": "USA",
  "email": "jane.doe@example.com",
  "emergencyContact": "John Doe - 555-1234",
}
```

```
"emergencyContact2": "Jane Smith - 555-5678",  
"allergies": "None",  
"notes": "Patient is in good health."  
}
```

Returns:

- Ok - response indicating the patient was successfully created.
- BadRequest - response if the provided patient information is invalid or incomplete.
- Unauthorized - response if the user does not have the necessary permissions to create a patient.
- InternalServerError - response if an unexpected error occurs during processing.

End Point: `api/getStudyFileUploadInfo`**Remarks:**

HTTP Method: POST

Description: Handles file upload actions including initiating a multipart upload, generating pre-signed URLs for multipart upload, completing a multipart upload, and obtaining a pre-signed URL for single file uploads. The method handles different types of actions based on the `actionType` parameter.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- `actionType`: The type of action to perform. Valid values are "start-multi", "get-multi-url", "complete-multi", and "single-upload".
- `studyID`: The ID of the study related to the file upload.
- `sDateTime`: The date and time of the file upload, formatted as "yyyyMMddHHmmss".
- `sFilename`: The name of the file to be uploaded.
- `partCount`: The number of parts for the multipart upload (used when `actionType` is "get-multi-url").
- `uploadId`: The upload ID for the multipart upload (used when `actionType` is "get-multi-url" and "complete-multi").
- `partETags`: The ETags for the uploaded parts (used when `actionType` is "complete-multi").

Returns:

- Ok - response with the upload ID if the multipart upload is successfully initiated, or the pre-signed URLs for multipart upload if `actionType` is "get-multi-url", or the single upload URL if `actionType` is "single-upload".

- NotFound - response if the `actionType` is invalid.
- BadRequest - response if the request parameters are invalid.
- Unauthorized - response if the authorization header is missing or invalid.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response for "start-multi":

```
{
  "UploadId": "example-upload-id"
}
```

End Point: api/getUploadedStudyFileSize

Remarks:

HTTP Method: GET

Parameters: FromBody Model

Description: Retrieves the size of an uploaded study file based on the provided study ID and filename. The file size is returned in bytes.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- info: An object containing the following parameters:

Example of request Body:

```
{
  "studyID": "1234abcd-5678-efgh-9101-1121ijklmno",
  "sDateTime": "20240912103000",
  "sFilename": "studydata.dcm",
}
```

Returns:

- Ok - response with the file size in bytes if the file exists and the request is successful.
- NotFound - response if the file does not exist or the provided parameters are invalid.
- Unauthorized - response if the authorization header is missing or invalid.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

```
{  
"Size": 1024  
}
```

End Point: `api/getDeviceStudySetup/{studyId}`

Remarks:

HTTP Method: GET

Parameters: string id

Description: Retrieves the device study setup information for a given study ID. See example for successful JSON response.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- studyId: The ID of the study for which to retrieve the device setup information. This should be a valid GUID string.

Returns:

- Ok - response with the device study setup information if the request is successful.
- Unauthorized - response if the authorization header is missing or invalid.
- Forbidden - response if the caller does not have the ViewStudy permission.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

```
{  
"n360url": "https://testserver.nexus360.cloud",  
"studyID": "12D192A1-A243-4C25-979A-D41F31160C0C",  
"patientID": "D625BFD0-4D0F-4F88-A2FD-ED86339B77AC",  
"serviceID": "BEC7621F-F769-4C6F-80C8-DE50F1E78E4B",  
"siteID": "AA81E334-DACC-4A59-B8D7-BF574B0FAD94",  
"reference": "CMP00001",  
"lastname": "Doe",  
"firstname": "John",  
"recordingMode": "Once",  
"recordingCount": "1",  
"recordingStart": "20240820220000",  
"recordingDuration": "28800",  
}
```

```
"studylocaltime": "20240819163116",
"settings": {
  "EXG1": "eeg",
  "EXG2": "emg",
  "EXG3": "eogl",
  "EXG4": "eogr",
  "BandsTestDuration": "3",
  "BandsRetryPeriod": "3600",
  "BandsRetryCount": "1"
}
```

End Point: `api/getDeviceStudySetupFromPIN/{pin}`

Remarks:

HTTP Method: GET

Parameters: string PIN

Description: Retrieves the device study setup information for a given PIN. See example for successful JSON response.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- studyId: The ID of the study for which to retrieve the device setup information. This should be a valid GUID string.

Returns:

- Ok - response with the device study setup information if the request is successful.
- Unauthorized - response if the authorization header is missing or invalid.
- Forbidden - response if the caller does not have the ViewStudy permission.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

```
{
  "n360url": "https://testserver.nexus360.cloud",
  "studyID": "12D192A1-A243-4C25-979A-D41F31160C0C",
  "patientID": "D625BFD0-4D0F-4F88-A2FD-ED86339B77AC",
  "serviceID": "BEC7621F-F769-4C6F-80C8-DE50F1E78E4B",
  "siteID": "AA81E334-DACC-4A59-B8D7-BF574B0FAD94",
```

```
"reference": "CMP00001",
"lastname": "Doe",
"firstname": "John",
"recordingMode": "Once",
"recordingCount": "1",
"recordingStart": "20240820220000",
"recordingDuration": "28800",
"studylocaltime": "20240819163116",
"settings": {
  "EXG1": "eeg",
  "EXG2": "emg",
  "EXG3": "eogl",
  "EXG4": "eogr",
  "BandsTestDuration": "3",
  "BandsRetryPeriod": "3600",
  "BandsRetryCount": "1"
}
```

End Point: `api/getPreparedDevicesx/{id}`

Remarks:

HTTP Method: GET

Parameters: string deviceSN

Description: Retrieves a list of prepared devices for a given device serial number. See example for successful JSON response.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- deviceSN: The serial number of the device for which to retrieve the list of prepared devices.

Returns:

- Ok - response with a list of prepared devices if the request is successful.
- NotFound - response if no prepared devices are found for the given serial number.
- Unauthorized - response if the authorization header is missing or invalid.
- Forbidden - response if the caller does not have the appropriate permissions.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

```
[
{
"deviceId": "1e9b1d85-4a4a-4b9b-b9e3-3e8aef67417a",
"deviceSN": "SN123456789",
"status": "Ready",
"preparedDate": "2024-09-10T12:34:56"
},
{
"deviceId": "5f9d3b1b-8a9f-4e5b-9a7b-5f82f5c2e9a9",
"deviceSN": "SN987654321",
"status": "In Use",
"preparedDate": "2024-09-11T08:15:22"
}
]
```

End Point: `api/getDeviceSites/{serialNumber}`

Remarks:

HTTP Method: GET

Parameters: string serialNumber

Description: retrieves a list of sites associated with a given device serial number.

Authentication Required: YES. Include Bearer TOKEN in the Header.

Parameters:

- serialNumber: The serial number of the device to search for associated sites.

Returns:

- A list of sites associated with the given device serial number.
- A `NotFoundResult` response if no sites are found for the given serial number.
- An `InternalServerErrorResult` response if an unexpected error occurs during processing.

Exceptions:

- `T:System.InvalidOperationException`: Thrown if an error occurs during data retrieval.
- `T:System.Exception`: Logs and rethrows any other exceptions encountered.

Example:

Example of a successful JSON response:

```
[
```



```
{  
  "siteID": "1111aaaa-2222-bbbb-3333-4444cccc5555",  
  "siteName": "Main Clinic"  
},  
{  
  "siteID": "6666dddd-7777-eeee-8888-9999ffff0000",  
  "siteName": "Secondary Clinic"  
}  
]
```

End Point: `api/assignDeviceToStudy/{type}/{deviceSN}/{studyID}`

Remarks:

HTTP Method: POST

Parameters: string type, string deviceSN, Guid studyID

Description: Assigns a device to a study. The device is specified by its type and serial number, and the study is identified by its ID. The API verifies permissions and existence of the device, study, and associated device group before performing the assignment.

Authentication Required: YES Bearer TOKEN in Header

Parameters:

- type: The type of the device being assigned. This should be a non-empty string that specifies the device type.
- deviceSN: The serial number of the device being assigned. This should be a non-empty string that uniquely identifies the device.
- studyID: The ID of the study to which the device will be assigned. This should be a valid GUID string representing the unique identifier of the study.

Returns:

- Ok - response with the details of the assigned device, study, and device group if the assignment is successful.
- BadRequest - response if the device type or serial number is empty.
- NotFound - response if the device, study, or device group is not found.
- Unauthorized - response if the authorization header is missing or invalid.
- Forbidden - response if the caller does not have the CreateStudy permission.
- InternalServerError - response if an unexpected error occurs during processing.

Example:

Example of a successful JSON response:

```
{  
  "DeviceID": "guid-of-the-device",  
  "StudyID": "guid-of-the-study",  
  "DeviceGroupID": "guid-of-the-device-group"  
}
```